

SOP-14: Low-Hazard Buffers, Salts, Stains, Detergents and Kits

*Roberts Lab, School of Aquatic and Fishery Sciences - Rev. 0 - September 1, 2026***PARTICULARLY HAZARDOUS SUBSTANCE (PHS): NO** - Sections 9, 10 and 11 are optional; they are completed below as good practice.

Section 1 - Lab-Specific Information

Building/Room(s) covered by this SOP	Fisheries Teaching & Research (FTR) Building, Rooms 209 and 213
Unit or department	School of Aquatic and Fishery Sciences (SAFS), College of the Environment
Principal Investigator	Steven Roberts (work phone 206-685-3742)
PI signature / date	_____ Date: _____
This SOP was created by (if not PI)	_____ Date: _____
Emergency contact	9-1-1 (campus phone) EH&S 206-543-7262 (M-F 8-5) 206-685-UWPD (8973) after hours
Date of last review	_____ Next review due: _____ (review annually and after any incident or process change)

Chemicals covered by this SOP

Chemical name (as listed in MyChem)	CAS number
1X Tris/EDTA (TE) solution pH 8.0; Tris(hydroxymethyl)aminomethane; Trizma HCl buffer solution	77-86-1
Ethylenediamine tetraacetic acid disodium salt dihydrate (EDTA)	6381-92-6
3M sodium acetate solution pH 5.2	127-09-3
Calcium chloride	10043-52-4
Sucrose for molecular biology	57-50-1
Agarose; agarose ultrapure	9012-36-6
Brilliant Blue R; Bromophenol blue; Trypan blue solution	6104-59-2 / 115-39-9 / 72-57-1
IGEPAL CA-630 (octylphenoxyethoxyethanol)	9002-93-1
DPBS; Dulbecco's PBS; phosphate buffered saline pH 7.4	mixture
RNAlater; DNA/RNA Shield (see also SOP-05)	mixture
pH 4.00 / 7.00 / 9.21 / 10.00 / 10.01 technical and color-coded buffer solutions	mixture
PAXgene Tissue DNA Kit, RNA Kit, Stabilizer Concentrate; Citrate Synthase Assay Kit	mixture

Safety Data Sheets for every chemical listed above must be readily available in the laboratory. SDSs are accessible through MyChem.

Section 2 - Hazards

Applicable GHS pictograms: Exclamation mark (eye and skin irritation) for some members; Environment (IGEPAL, aquatic toxicity Cat 1); most items carry no GHS pictogram

- This SOP covers the reagents in the Roberts Lab inventory that MyChem records as 'No identified hazard at this time', 'No identified hazards at this time', 'NOT CLASSIFIED', or as irritants only. It exists because UW requires that all chemicals in a laboratory be addressed by an SOP, and because several items in this group carry real if modest hazards that are easy to overlook.
- Stock chemicals and working solutions: buffers and salts used for nucleic acid work (TE, Tris, EDTA, sodium acetate, calcium chloride, sucrose), agarose for gel electrophoresis, protein and viability stains (Brilliant Blue R, bromophenol blue, trypan blue), the non-ionic detergent IGEPAL CA-630, physiological salines (DPBS, PBS), sample preservatives (RNAlater, DNA/RNA Shield), commercial pH calibration buffers, and prepackaged assay and extraction kits. Wastes: spent buffers, stained gels (unstained by ethidium bromide), used preservative, and empty kit components.
- The items in this group that deserve specific attention are: IGEPAL CA-630, which causes serious eye damage (Category 1) and is very toxic to aquatic life (acute and chronic Category 1); trypan blue, which is a suspected carcinogen and teratogen in some classifications despite the sparse MyChem record; calcium chloride and EDTA, which are eye irritants; sucrose and agarose powders, which are combustible dusts in bulk; and the kits, whose individual buffer components may be hazardous even when the kit as a whole is recorded as 'NOT CLASSIFIED'.
- Routes of exposure: eye contact (the main concern), skin contact, inhalation of dust when weighing powders, ingestion.
- How exposure might occur: weighing powders, splashing buffer while pipetting or pouring, contact with molten agarose (a thermal burn hazard at 90-100 degC), and handling kit reagents whose composition is not obvious from the box.
- Target organs: eyes and skin, primarily. Molten agarose causes thermal burns. Superheated agarose in a microwave can boil over violently when the flask is moved.
- Signs and symptoms of exposure: eye redness, watering and irritation; transient skin irritation or dryness; a thermal burn from molten agarose.
- GHS classification (from MyChem / SDS): most items in this group - 'No identified hazards at this time' or not classified. IGEPAL CA-630 - Acute toxicity Category 4 (oral); Serious eye damage Category 1; Skin corrosion/irritation Category 2; Acute and chronic aquatic toxicity Category 1. Calcium chloride and 'agarose, ultrapure' - eye irritation Category 2A / irritant. EDTA disodium - irritant. pH 7.00 buffer (910107) - Skin corrosion/irritation Category 2; STOT-repeated Category 2; eye irritation Category 2. Sucrose - combustible dust. Buffer pH 4.00 - hazard not otherwise classified.
- IMPORTANT LIMITATION: an entry of 'NOT CLASSIFIED' or 'No identified hazard at this time' in MyChem means the hazard field has not been populated, not that the manufacturer has determined the product is non-hazardous. Before first use of any kit or reagent in this group, the user must read the current manufacturer SDS. If the SDS shows any GHS health or physical hazard category, that item is removed from this SOP and covered by the appropriate hazard-class SOP instead, and the PI is notified.

Section 3 - Engineering Controls and Personal Protective Equipment (PPE)

Engineering controls

- No special engineering controls are required for the routine handling of these reagents. General laboratory ventilation is adequate.

- Weighing bulk powders (agarose, sucrose, Tris, EDTA, calcium chloride) is done at the bench with reasonable care to avoid raising dust; if a large quantity (more than approximately 100 g) is weighed, do it in the fume hood or use a balance enclosure to control dust. Any chemical fume hood used must be tested and passed by EH&S.
- IGEPAL CA-630 is dispensed in the certified chemical fume hood in FTR 209 because it causes serious eye damage and is a viscous liquid that is prone to dripping and stringing.
- Melting agarose: use a microwave with the flask loosely covered and no more than half full, heat in short intervals with swirling between them, and let the flask stand for 30 seconds before removing it. Superheated agarose erupts. Use heat-resistant gloves and hold the flask away from the body.
- Kit reagents are handled per the manufacturer's instructions; any kit component identified in the SDS as flammable, corrosive, acutely toxic or guanidinium-containing is handled under the applicable SOP (SOP-01, SOP-05 or SOP-07) rather than this one.
- The nearest eyewash and safety shower are located in FTR 209 and FTR 213 and are inspected and flushed weekly by lab staff.

Hygiene measures

- Avoid contact with skin, eyes, and clothing. Wash hands after removing PPE, before breaks, and immediately after handling the chemical. If any chemical covered by this SOP comes into contact with any PPE, the PPE shall be immediately removed and discarded properly. Any potentially exposed body parts should be washed immediately.
- No eating, drinking, chewing gum, applying cosmetics, or storing food or drink in any area where these chemicals are used or stored.

Skin and body protection

- Chemically compatible laboratory coats that fully extend to the wrist must be worn, appropriately sized and buttoned to their full length. Personnel must also wear full-length pants, or equivalent, and close-toe shoes. The area of skin between the shoe and ankle must not be exposed.
- Lab coat type: standard buttoned barrier lab coat is sufficient for all work in this SOP.
- Heat-resistant gloves are worn when handling flasks of molten agarose.
- No additional chemically resistant clothing is required, except that disposable sleeves are worn when dispensing IGEPAL CA-630.

Hand protection

- Hand protection IS required for the activities described in this SOP.
- Required gloves: 4-6 mil nitrile examination gloves for all handling of these reagents, changed on any visible contact. Nitrile is compatible with all aqueous buffers, salts, stains and detergents in this group.
- Heat-resistant or insulated gloves are worn over nitrile when handling molten agarose.
- Consult your preferred glove manufacturer to ensure that the gloves you plan to use are compatible with the specific chemical being used. Gloves must be inspected prior to use, including a check for pinholes.
- Use proper glove removal technique (without touching the glove's outer surface) to avoid skin contact with this product. Dispose of contaminated gloves after use in accordance with applicable laws and good laboratory practices. Wash and dry hands immediately after glove removal.

Eye protection

- ANSI Z87.1-compliant eye protection IS required for all work with these chemicals. Ordinary prescription glasses will NOT provide adequate protection unless they also meet the Z87.1 standard and have compliant side shields.

- Minimum eye protection: ANSI Z87.1 safety glasses with side shields for all routine work with these reagents. Chemical splash goggles are required when dispensing IGEPAL CA-630 (serious eye damage Category 1), when pouring molten agarose, and when handling any volume greater than 1 L.

Respiratory protection

- Respiratory protection is NOT required for the activities described in this SOP.
- All reagents in this group may be used outside of the fume hood at any laboratory bench, with the exception of IGEPAL CA-630 dispensing and of bulk powder weighing above approximately 100 g, both of which are done in the fume hood.
- Respirators should be used as a last line of defense (i.e., after engineering and administrative controls have been exhausted), and when any Action Limit (AL) or Occupational Exposure Limit (OEL) has been exceeded or when there is a possibility that an AL/OEL will be exceeded. If a potential exposure hazard cannot be eliminated, contact the EH&S Respiratory Protection Program administrator to discuss respiratory protection or to enroll in the program so a respiratory protection analysis can be performed. Program enrollment includes medical evaluation, training and fit testing for an appropriate respirator.

Section 4 - Special Handling and Storage Requirements

- Read the SDS before first use of any reagent or kit in this group. If the SDS discloses a GHS hazard category not reflected in MyChem, stop, notify the PI, and use the applicable hazard-class SOP instead.
- IGEPAL CA-630 is very toxic to aquatic life (Category 1). Do not pour it, or buffers containing it, down the drain or into any seawater system; collect it as chemical waste.
- Trypan blue is handled with gloves and safety glasses and is collected as chemical waste rather than sinked, because some classifications treat it as a suspected carcinogen and teratogen and the MyChem record for it is incomplete.
- Molten agarose: never carry an uncovered flask of near-boiling agarose across the lab; never microwave a tightly capped bottle; never reheat a bottle that has been capped while hot without loosening the cap first.
- Bulk sucrose and agarose powders are combustible dusts; do not create dust clouds and keep them away from ignition sources.
- All prepared solutions must be labeled with the full chemical name (not an abbreviation alone), the concentration, the date and the preparer's initials. 'TE' and 'PBS' alone are not adequate labels for a bottle that will outlive the person who made it.
- Storage: these reagents are stored on general laboratory shelving and in the refrigerator or freezer as the manufacturer directs, in FTR 209 and FTR 213. Segregate from strong acids, strong bases and oxidizers. Secondary containment is used for liquids larger than 1 L. Nothing in this group requires a locked or restricted cabinet.
- Transport: containers larger than 1 L are carried in a bottle carrier.
- Unattended operations: gels running and samples incubating in closed vessels may be left unattended with appropriate signage.
- Clean work surfaces upon completion of tasks with 70% ethanol and wipe dry.
- Clean all contaminated surfaces with detergent solution or 70% ethanol and dry.
- Place all contaminated disposable items in appropriate laboratory waste for disposal.
- Non-disposable/re-usable utensils, glassware, and other surfaces contaminated with these reagents are washed with detergent and water and rinsed with deionized water at the end of the laboratory work session.
- When work is completed, remove gloves and wash hands with soap and water.

Section 5 - Spill and Accident Procedures

Spill response

- Chemical spills must be cleaned up as soon as possible by properly protected and trained personnel. All other persons should leave the area. Do not attempt to clean up any spill if not trained or comfortable. Evacuate the area and call 9-1-1 on a campus phone for help.
- Small spill - up to approximately 1 L of any aqueous buffer, salt, saline, preservative or stain solution in this group: may be handled by trained lab staff wearing safety glasses, nitrile gloves and a lab coat. Absorb with paper towels or absorbent pads working from the outside inward, place in a plastic bag, wipe the area with detergent solution followed by 70% ethanol, and dry.
- Powder spill (agarose, sucrose, Tris, EDTA, calcium chloride): do not blow or brush it into the air; sweep gently with a damp wipe or dampen and wipe up, working from the outside inward. Place in a plastic bag.
- IGEPAL CA-630 spill of any size: treat as a hazardous spill. It causes serious eye damage and is very toxic to aquatic life. Wearing goggles and nitrile gloves, absorb with absorbent pads, place in a screw-top hazardous waste container, and wash the area with detergent and water. Do not flush to a drain. Report any IGEPAL spill that reaches a drain to EH&S.
- Molten agarose spill: do not touch it. Allow it to cool and solidify, then lift it off the surface and discard it as solid waste. Treat any skin contact as a thermal burn - cool with running water for 20 minutes and seek medical evaluation.
- Any spill greater than approximately 1 L, or any spill in which a person is injured, or any spill of unknown kit reagent: restrict access, post a sign, and contact EH&S at 206-543-0467 during business hours or via 9-1-1 after hours.
- PPE required for cleanup: safety glasses (goggles for IGEPAL), nitrile gloves, buttoned lab coat, full-length pants and close-toe shoes.
- Spill area must be cleaned up in the following manner: after absorbent removal, clean the spill area thoroughly with detergent solution followed by clean water, then 70% ethanol, and dry.
- Spill cleanup materials must be disposed of in the following manner: absorbent from non-hazardous aqueous buffer spills may be discarded as regular laboratory solid waste once dry. Absorbent from IGEPAL, trypan blue or unknown kit reagent spills is double bagged, placed in a screw-top container, labeled with a UW Hazardous Waste Label, and submitted to EH&S for collection.
- For questions on spill cleanup, contact EH&S spill consultants at 206-543-0467 during normal business hours (Monday-Friday, 8 a.m. to 5 p.m.).
- Any spill, exposure or near miss incident requires the involved person or supervisor to complete and submit an incident report on the EH&S website within 24 hours (certain types of incidents require immediate notification).

Exposure procedures - first aid

- Perform first aid immediately. Refer to the SDS for additional chemical-specific guidance.
- Eye exposure: use the eyewash for 15 minutes while holding the eyelids open. For IGEPAL CA-630, which causes serious eye damage, continue for 15 minutes and go immediately to the Emergency Department.
- Skin exposure: wash the affected area with soap and running water for 15 minutes; remove contaminated clothing.
- Thermal burn from molten agarose: cool the burn under cool running water for at least 20 minutes, do not apply ice or ointment, and seek medical evaluation for anything larger than a small superficial burn.
- Inhalation exposure: move the person to fresh air; get medical help if irritation persists.
- Ingestion: rinse mouth with water; get medical help.
- Sharps injury (needle stick or subcutaneous exposure): scrub the exposed area thoroughly for 15 minutes using warm water and sudsing soap.

- Get help. Call 9-1-1 or go to the nearest Emergency Department if the injury is more than minor; provide details of exposure: agent, dose, route of exposure, and time since exposure. Bring the SDS and this SOP to the Emergency Department.
- Notify your supervisor as soon as possible for assistance.
- Notify EH&S immediately after providing first aid and/or getting help. During business hours (M-F, 8-5) call 206-543-7262. Outside of business hours call 206-685-UWPD (8973) to be routed to EH&S staff on call.

Section 6 - Waste Accumulation and Disposal Procedures

- Waste streams: (a) non-hazardous aqueous buffers, salines and salt solutions; (b) IGEPA CA-630 and detergent-containing buffers; (c) stains (trypan blue, Brilliant Blue R, bromophenol blue); (d) agarose gels not stained with ethidium bromide; (e) spent preservatives; (f) kit reagent waste; (g) solid debris.
- Sewer disposal: dilute, non-hazardous aqueous buffers and salines (TE, Tris, PBS, DPBS, sodium acetate, calcium chloride, sucrose, commercial pH calibration buffers) that contain no other hazardous constituent may be disposed of to the sanitary sewer with copious running water, consistent with EH&S chemical treatment and recycling guidance. Confirm against the SDS before sewerage of any specific solution, and never sewer a solution that has been used with a hazardous chemical from another SOP.
- IGEPA CA-630 and any buffer containing it must NOT be sewerage - it is very toxic to aquatic life at Category 1. Collect it as hazardous waste.
- Trypan blue and other stain solutions are collected as hazardous waste rather than sewerage, given incomplete hazard records.
- Agarose gels without ethidium bromide are discarded as regular laboratory solid waste once solidified. Gels stained with ethidium bromide are hazardous waste under SOP-06.
- Kit reagent waste: route by the SDS of the specific component. Guanidinium-containing kit buffers go to the SOP-05 guanidinium waste stream and must never be acidified or bleached; alcohol-containing wash buffers go to the SOP-01 solvent waste stream. Do not assume a kit component is non-hazardous because the kit is recorded as NOT CLASSIFIED.
- Accumulate any hazardous liquid waste at the point of generation in a sturdy, chemically compatible container with a securely closable screw-top lid, kept closed except when actively adding waste, in secondary containment.
- All chemical waste containers must be labeled with a UW Hazardous Waste Label listing every constituent and approximate percentage, starting from the moment the first drop of waste is added. Refer to 'How to Label Chemical Waste Containers' on the EH&S website.
- Manage chemical and hazardous chemical waste separately from other waste streams such as biohazardous waste. Never autoclave chemical waste because it can produce hazardous chemical vapors, aerosols, and explosive reactions.
- In certain cases, chemical waste can be treated and disposed of into the sanitary sewer or exchanged with other University units. Chemical treatment and recycling and chemical exchange options are available on the EH&S website.
- The University of Washington Environmental Health & Safety Department has full responsibility for collection of hazardous waste for the University, all its campuses, and off-site locations; University laboratories cannot contract with an outside vendor to collect hazardous waste.
- To request a collection of chemical waste, submit a form on the Chemical Waste Disposal webpage on the EH&S website or directly in MyChem inventory. Contact EH&S at 206-616-5835 or chmwaste@uw.edu with questions.
- The inventory contains several duplicate and long-held containers in this group (two agarose containers, two DPBS containers, multiple pH buffers). Review these annually and surplus or exchange what is not needed.

- Visit the Hazardous Material Disposal and Recycling webpage on the EH&S website for information on disposing, recycling and surplusing materials. Update the MyChem inventory when containers are emptied, received, transferred or surplused.

Section 7 - Protocol

- Procedure A - Preparing buffers and salt solutions. Weigh the solid at the bench with nitrile gloves and safety glasses, dissolve in deionized water with stirring, adjust pH with dilute acid or base per SOP-07 and SOP-08, bring to volume, and label with the full chemical name, concentration, pH, date and initials. Autoclave or filter-sterilize as the protocol requires.
- Procedure B - Casting agarose gels (without ethidium bromide). Weigh agarose, add buffer, and melt in a microwave in short intervals with swirling, using a flask no more than half full and loosely covered. Wear heat-resistant gloves and safety glasses. Cool to approximately 55-60 degC before adding any stain and pouring. For ethidium bromide-stained gels, follow SOP-06.
- Procedure C - Dispensing IGEPAL CA-630. Inside the certified fume hood, wearing goggles, nitrile gloves, lab coat and disposable sleeves, dispense the viscous detergent with a positive-displacement pipette or by weight. Wipe the bottle neck before recapping. Collect all IGEPAL-containing waste separately.
- Procedure D - Sample preservation in RNAlater or DNA/RNA Shield. Aliquot preservative into labeled tubes, add tissue, cap securely, and store as the protocol directs. DNA/RNA Shield is also covered by SOP-05; the more protective requirement governs.
- Procedure E - Commercial kits (PAXgene, citrate synthase assay, pH calibration buffers). Follow the manufacturer's instructions. Before first use, read the SDS for each kit component and confirm that no component requires a different SOP.
- Refer to Section 2 of the UW Laboratory Safety Manual on the EH&S website for additional guidance on chemical management and preparation for use.
- NOTE: Any deviation from this SOP requires approval from the Principal Investigator.

Section 8 - Special Precautions for Animal Use

No. N/A - buffers and salines in this group may be used to rinse or hold tissue and, in the case of physiological salines, to maintain live animals under an IACUC-approved protocol. None of these reagents is administered as a test substance. IGEPAL CA-630 and trypan blue are not used with live animals.

IACUC protocol number (if applicable): _____

PARTICULARLY HAZARDOUS SUBSTANCE (PHS): NO - Sections 9, 10 and 11 are optional; they are completed below as good practice.

Section 9 - Approvals Required

- All staff working with the reagents covered by this SOP must be trained on this SOP prior to starting work. They must also review the SDS for each reagent, and the SDSs must be readily available in the laboratory (MyChem). All training must be documented and maintained by the PI or their designee.
- Prerequisite training, documented before independent work: (1) EH&S Laboratory Safety Training, initial and current refresher; (2) EH&S Managing Laboratory Chemical Waste training; (3) documented review of this SOP.

- No PI authorization beyond general laboratory access is required for the reagents in this group, with two exceptions: dispensing IGEPAL CA-630, and any first use of a kit or reagent whose SDS has not previously been reviewed by the lab, both of which require the user to confirm with the PI or lab manager that the correct SOP is being applied.
- A worker must have documented hands-on training from the PI or a designated senior lab member on agarose melting and pouring before doing it independently, because the thermal burn hazard is the most likely injury in this group.
- No medical examination or respirator authorization is required for the work described.
- The PI should ask EH&S to update the MyChem hazard records for the items in this group currently listed as NOT CLASSIFIED or with empty GHS fields, so that the inventory reflects the manufacturer SDS.

Section 10 - Decontamination

- Work surfaces: wipe with detergent solution or 70% ethanol and dry at the end of each session.
- Equipment: wipe pipettors, racks, gel boxes and balances with 70% ethanol. Balance pans used for powders are brushed clean with a dedicated brush and wiped.
- Glassware: wash with detergent and hot water and rinse three times with deionized water. Glassware used for IGEPAL is rinsed first into the IGEPAL waste container.
- Microwave used for agarose: wipe out spills promptly; dried agarose is removed with hot water.
- Controlled areas: none of the reagents in this group requires a designated area or a controlled work zone.
- Cleaning solutions and materials: laboratory detergent (Alconox or Contrad) solution, hot water, deionized water, 70% ethanol, lint-free wipes.

Section 11 - Designated Area

- No designated area is required for the reagents in this group, because none of them is a particularly hazardous substance. Sections 9-11 are provided for completeness and are optional for this SOP.
- Work locations: any laboratory bench in FTR 209 and FTR 213, with the exceptions below.
- IGEPAL CA-630 dispensing: the certified chemical fume hood in FTR 209.
- Bulk powder weighing above approximately 100 g: the certified chemical fume hood in FTR 209 or a balance enclosure.
- Agarose melting and pouring: the designated gel bench, next to the microwave, with heat-resistant gloves available.

Section 12 - Documentation of Training

- Prior to using substances included in this SOP, laboratory personnel must be trained on the hazards described in this SOP, how to protect themselves from the hazards, and emergency procedures.
- Ready access to this SOP and to a Safety Data Sheet for each hazardous material described in the SOP must be made available in the lab space(s) where these substances are used.
- The Principal Investigator, or Responsible Party if the activity does not involve a PI, must ensure that their laboratory personnel have attended appropriate laboratory safety training (and refresher training where applicable).
- Training must be repeated following any revision to the content of this SOP.
- Training must be documented. This training sheet is provided as one option; other forms of training documentation (including electronic) are acceptable but records must be accessible and immediately available upon request.

I have read and understand the content of this SOP:

[illegible]